

RUBY_ESSENTIALS

Ruby, A Programmer's Best Friend

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Disclaimer: The content presented here is a curated blend of my personal learning journey, experiences, open-source documentation, and invaluable knowledge gained from diverse sources. I do not claim sole ownership over all the material; this is a community-driven effort to learn, share, and grow together.

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Overview

Ruby: Interpreted, dynamic, object-oriented scripting language.

Created By: Matz 1993

Inspiration: Combines Python's simplicity with Smalltalk's true OOP.

Rise in Popularity

Boom: After David Heinemeier Hansson (DHH) released Ruby on Rails (Rails) in 2005.

Why Popular?:

- Rails = full-stack MVC web framework.
- Helped startups rapidly build web apps.
- Used By: Twitter, Shopify, GitHub, Airbnb.

What is an Object in OOP (Object-Oriented Programming)?

An object is a real-world entity or instance of a class that contains:

- Data (called attributes or properties)
- Behavior (defined through methods or functions)

MVC: Model–View–Controller

MVC is a software design pattern commonly used for developing user interfaces that divide an application into three interconnected components:

Components of MVC

Component	Responsibility
Model	Handles data, business logic, and rules of the application.
View	Manages the UI – what the user sees.
Controller	Acts as the intermediary between Model and View – handles user input, processes it, and updates Model/View accordingly.

Package Management

RubyGems: The official package manager.

Gem: Self-contained library/module/package.

Setup

```
sudo apt install ruby
ruby -v      # Should print something like: ruby 3.x.x
```

```
nano file.rb
ruby file.rb
```

Takeaways

- Simple, English-like syntax
- Everything is an object
- Full OOP + functional flexibility
- Duck typing allows polymorphism without inheritance
- Rails made it legendary for web apps
- Huge ecosystem with RubyGems

ruby.rb

- install ruby `sudo apt-get install ruby-full`
- save file in extension `file.rb`
- run command `ruby file.rb`

```
# extension: file.rb
# run:ruby file.rb
# comment

# ruby is true OPPS lang
# dynamically typed
# everything is an object and every object can be modified

# var, now datatype declare
name = "Bali"
$age = 25 # $global_var
# CONST
LANG = "Ruby"
# $GlobalVar
$GlobalVar = "3.0"

# each stuff is an object
# thus object methods play is possible
puts "hello".upcase() # () parentheses are optional
LANG = LANG.upcase #this works but with warning as re-assinging constants
is not supposed norm,warning: already initialized constant LANG

puts LANG

# method declare
def greet(name)
  "Hello, #{name}"
end

puts greet("Bhati")
```

```
name = "robin"
puts "Hi, my name is #{name.upcase}"
puts "2+2=#{2+2}"
print " ,okay :)\n"

# print, print without \n
# puts , print + \n, puts (short for "put string")
# return, return obj

dt1=-14 #int
dt2=-14.2 #float
dt3="String"#string
dt4=true#boolean
dt5=nil#null
dt6=[1,2,1.2]#array
dt7={id:"2123"}#hash
dt8=:lightstring #lightstring

puts dt1.class
puts dt2.class
puts dt3.class
puts dt4.class
puts dt5.class
puts dt6.class
puts dt7.class
puts dt8.class
# Integer
# Float
# String
# TrueClass
# NilClass
# Array
# Hash
# Symbol

# Everything is a String. A single character, such as 'A', is simply a
String object that happens to have a length of one. Ruby does not have a
separate, distinct character data type like the char type found in
languages like C or Java.

#string methods
name = "Ruby"
puts name.downcase
puts name.upcase #upcase, not uppercase
puts name.class
puts name.length
puts name.reverse
puts name.include?("B")
# ruby
# RUBY
# String
# 4
```

```
# ybuR
# false

# numbers
a= 3
b = 16
puts a + b # 19
# + - * / % .to_f
a=a+b/(a*b)
puts a.to_f #3.0

# boolean
puts true && false # and
puts true || false # or
puts !true # not
# false
# true
# false

# Array
fruits = ["apple", "banana", "mango"]
puts fruits
# apple
# banana
# mango
puts fruits[0] # apple
fruits.each {|n| puts n.upcase() }
# APPLE
# BANANA
# MANGO

# Hashes
user = { name: "Alice", age: 25 }
puts user # {:name=>"Alice", :age=>25}
puts user[:name] #Alice
puts user.keys
# name
# age

# conditions
# ruby uses if,else,elsif(not else if or elif) like python but req. 'end'
# to close
# ruby is indented language
age = 18
if age >=18
  puts "YouAreAdult"
elsif age == 17
  puts "WaitOneYear"
else
  puts "Minor"
end

# .between?()
```

```
# if shortcut
puts "TEen" if age.between?(13,19)
# in Ruby, the .between?() method includes both the lower limit and the
upper limit in its check.

# comparison operators
# == != > < <= >=

# Loops in ruby
# 1 while loop
# do till its true
i = 1
while i<=5
  puts "Counter #{i}"
  i+=1# not i++
end

# 2 Until loop
# do untill this is true
# i.e. do till only false
until i>10
  puts "Counter #{i}"
  i+=1
end

# 3 times loop
10.times do |i| # i is var inside times loop (default 0)
  puts "TImes #{i}"
end

puts "TImes #{i}"

# 4 for loop
for i in 1..15
  puts "#{i}"
end

# 5 each loop
arr1=[1,2,4]
arr1.each do |e|
  puts "arr1 #{e}"
end

# Methods in Ruby
# functions
# def -- end
# return only one object
# they return last evaluated (explicit return is optional)

def greet(name="Bali") # arguement and its default setup
  # This method takes one argument and returns a greeting string
  puts "Hello, #{name}!"
end
```

```
greet("Bhati")
greet()

def sum(n,n2)
  return n+n2 # Optional; could omit `return`
end
puts sum(1,2)

# methods with conditions
def even_or_odd(num)
  if num % 2 == 0
    "Even"
  else
    "Odd"
  end
end

puts even_or_odd(10) # Even
puts even_or_odd(7) # Odd

# Blocks in ruby
# anonymous chunks of code you can pass to methods
# written do ... end or { ... }
# anonymous: not stored in a variable or named method.

# each with do ... end block
[1,2,4].each do |n| # |pipe cahracter are arguements starting from 0|
  puts n
end

# same with { ... } block
[5,6,8].each{ |n| puts n }

# yield = putting statement as arguement , can be repetitive in method
# yield is a keyworld used inside a method to call a block that was passed
implicitly

def greet
  puts "Before yield"
  yield
  puts "After yield"
end

greet { puts "STATEMENT AS ARGUMENT " }

# multiple calls
def twice
  yield
  yield
end
```

```
twice { puts "Run Block" }

# block with arguments
def food_time
  yield("Pizza", 2)
end

food_time do |food,qty|
  puts "#{qty} #{food}s" # 2 Pizzas
end

# Symbols
# immutable identifier used as name or label
# begins with colon :eg

:admin
:email
:username
:token

# Symbols are designed as immutable, unique, internal identifiers.

# Strings are designed as mutable sequences of characters for storing and
manipulating text.

# symbols are memory efficient than strings when reused
puts :admin == :admin #true
puts "admin" == "admin" #true
# but symbols always point to same internal object
puts :admin.object_id
puts :admin.object_id
puts :admin.object_id
puts "admin".object_id
puts "admin".object_id
puts "admin".object_id
# 1355868
# 1355868
# 1355868
# 60
# 80
# 100

# keyword args in methods
# Instead of passing args by position, Ruby lets you name them (like
Python)

def creatuser(name:,age:) # here its name: (not :name)
  puts "Created #{name}, age #{age}" # here name (not name:)
end

creatuser(name:"Bob",age:22) #here name:
# Created Bob, age 22
```



```
# OOPS basics
# Classes and Objects
# class
# blueprint for creating objs with shared behaviour
class User
  # attr_accessor creates both getter and setter methods automatically
  # this gives access to @name and @email from outside the object
  attr_accessor :name, :email

  # initialize method
  # run when you do User.new(...)
  # generates instance variables like @name, @email

  def initialize(name, email)
    @name = name #instance variable for this object
    @email = email
  end

  # instance is a single object created from class
  # class = blueprint = recipe
  # instance = actual object made from it = cake
  # instance method

  def greet
    "Hi, i am #{@name}"
  end
end

# create an obj
user1 = User.new("Siddhant Bali", "kintsugidevstudio@gmail.com")

# Call instance methods and accessors
puts user1.greet
puts user1.name
puts user1.email
# Hi, i am Siddhant Bali
# Siddhant Bali
# kintsugidevstudio@gmail.com

# Changing values using setter
user1.name = "Siddhant Bali"
puts user1 # =<User:0x0000735b55c97880>
puts user1.name
puts user1.greet
# Siddhant Bali
# Hi, i am Siddhant Bali

# Modules
# container of reusable methods
# can't be initialized like class
# You "include" a module in a class to add its behavior (mixin)
module Printer
  def puts
```

```
    print "Name #{@name}\n"
  end
end

# Inheritance
# classes can inherit from others classes using '<'
# inheriting allows to reuse code logic defined in parent class

# parent class
class Employee
  attr_accessor :name, :id

  def initialize(name, id)
    @name = name
    @id = id
  end

  def greet
    "Welcome #{@name}, Your Id is #{@id}"
  end
end

class Admin < Employee
  include Printer # Mixin: Adds methods from module as instance methods
end

# create object
admin1 = Admin.new("Bali", 2022496)
puts admin1.greet
admin1.puts
# Welcome Bali, Your Id is 2022496
# Name Bali
```

Ruby Concepts + Detailed Code Explanation

1. Introduction to Ruby Programming

- Ruby is an Object-Oriented Programming Language (OOP).
- It is dynamically typed, meaning variable types are inferred at runtime, not at compile-time.
- Everything in Ruby is an object, and every object can be modified.

```
# Variable Declaration
name = "Bali" # String variable
$age = 25     # Global variable
```

- Global Variables: `$age` is a global variable, meaning it is accessible throughout the program.

```
# print, print without \n
# puts , print + \n, puts (short for "put string")
# return, return obj
```

2. Constants

- Constants in Ruby are declared using uppercase names and should not be reassigned.

```
# Constants in Ruby
LANG = "Ruby"           # Constant
$GlobalVar = "3.0"      # Another global variable (not a constant)
```

- Warning: If a constant is reassigned, Ruby will give a warning.

3. Methods in Ruby

- Methods are declared using `def` and closed with `end`.
- They can return a single object (implicitly or explicitly).
- Parameters can have default values.

```
# Method with a parameter
def greet(name)
  "Hello, #{name}"
end

puts greet("Bhati") # Output: Hello, Bhati
```

- Default Values in Methods: If no argument is passed, a default value can be used.

```
def greet(name="Bali")
  puts "Hello, #{name}!"
end

greet("Bhati") # Output: Hello, Bhati
greet()       # Output: Hello, Bali
```

4. String Manipulation Methods

- Ruby has various built-in string methods for manipulation.

```
# String Methods
name = "Ruby"
puts name.downcase # ruby
puts name.upcase   # RUBY
puts name.class     # String
puts name.length   # 4
puts name.reverse   # ybuR
puts name.include?("B") # false
```

- Explanation:
 - **downcase**, **upcase**: Convert the string to lowercase/uppercase.
 - **reverse**: Reverses the string.
 - **include?**: Checks if a substring exists in the string.

Everything is a String. A single character, such as 'A', is simply a String object that happens to have a length of one. Ruby does not have a separate, distinct character data type like the char type found in languages like C or Java.

5. Numbers in Ruby

- Ruby supports basic arithmetic operations and conversion between data types.

```
# Numbers in Ruby
a = 3
b = 16
puts a + b           # 19
puts (a + b) / (a * b) # Fraction result
puts a.to_f          # 3.0 (convert to float)
```

6. Boolean Operations

- Ruby provides boolean logic (**and**, **or**, **not**).

```
# Boolean Operations
puts true && false # false (AND)
puts true || false # true (OR)
puts !true         # false (NOT)
```

7. Arrays in Ruby

- Arrays are ordered collections that can hold multiple data types.

```
# Arrays in Ruby
fruits = ["apple", "banana", "mango"]
puts fruits
puts fruits[0]          # apple
fruits.each {|n| puts n.upcase} # APPLE, BANANA, MANGO
```

8. Hashes in Ruby

- Hashes are key-value pairs.

```
# Hash in Ruby
user = { name: "Alice", age: 25 }
puts user          # {:name=>"Alice", :age=>25}
puts user[:name]    # Alice
puts user.keys      # [:name, :age]
```

9. Conditional Statements

- Ruby uses `if`, `elsif`, `else`, and `end`.

```
# Conditional Statements
age = 18
if age >= 18
  puts "You are an adult"
elsif age == 17
  puts "Wait one year"
else
  puts "Minor"
end
```

- Shortcut for conditions: You can use `between?` to check ranges.

```
# If shortcut
puts "Teen" if age.between?(13, 19) # Output: Teen
```

in Ruby, the `.between?()` method includes both the lower limit and the upper limit in its check.

10. Loops in Ruby

- Ruby supports multiple loop types:

1. While Loop

```
i = 1
while i <= 5
  puts "Counter #{i}"
  i += 1
end
```

2. Until Loop

```
i = 1
until i > 10
  puts "Counter #{i}"
  i += 1
end
```

3. Times Loop

```
10.times do |i|
  puts "Times #{i}"
end
```

4. For Loop

```
for i in 1..15
  puts "#{i}"
end
```

5. Each Loop

```
arr1 = [1, 2, 4]
arr1.each do |e|
  puts "arr1 #{e}"
end
```

11. Defining Blocks

- Blocks are anonymous functions passed to methods.

```
# Each loop with a block
[5, 6, 8].each { |n| puts n }
```

```
# Yielding inside a method
def greet
  puts "Before yield"
  yield
  puts "After yield"
end

greet { puts "Hello from the block!" }
```

12. Symbols

- Symbols are lightweight strings used for identifiers.

```
# Symbols in Ruby
puts :admin == :admin # true
puts "admin" == "admin" # true
```

- Symbols are memory efficient and point to the same internal object.

Symbols are designed as immutable, unique, internal identifiers.

Strings are designed as mutable sequences of characters for storing and manipulating text.

13. Keyword Arguments in Methods

- You can define named arguments in Ruby methods.

```
# Keyword arguments
def create_user(name:, age:)
  puts "Created #{name}, age #{age}"
end

create_user(name: "Bob", age: 22)
```

14. Classes and Objects

- Classes define the blueprint for objects. Objects are instances of a class.

```
# Class Definition
class User
  attr_accessor :name, :email

  def initialize(name, email)
```

```
@name = name
@email = email
end

def greet
  "Hi, I am #{@name}"
end
end

# Creating an object
user1 = User.new("Siddhant Bali", "kintsugidevstudio@gmail.com")

# Calling methods
puts user1.greet      # Hi, I am Siddhant Bali
puts user1.name       # Siddhant Bali
```

- **attr_accessor**: Automatically creates getter and setter methods for instance variables.

15. Inheritance in Ruby

- Ruby allows classes to inherit from other classes, enabling reuse of code.

```
# Parent Class
class Employee
  attr_accessor :name, :id

  def initialize(name, id)
    @name = name
    @id = id
  end

  def greet
    "Welcome #{@name}, Your Id is #{@id}"
  end
end

# Child Class Inheritance
class Admin < Employee
  include Printer
end

admin1 = Admin.new("Bali", 2022496)
puts admin1.greet      # Welcome Bali, Your Id is 2022496
admin1.print_name      # Name Bali
```

- Inheritance: **Admin** class inherits from **Employee**, reusing **greet** and adding **print_name** functionality through **Printer** module.

16. Modules in Ruby

- Modules provide reusable methods that can be mixed into classes.

```
# Defining a module
module Printer
  def print_name
    puts "Name #{@name}"
  end
end

# Including a module
class Admin
  include Printer
end

admin = Admin.new("Bali", 2022496)
admin.print_name # Name Bali
```

End-of-File

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